

Seminar on Control Theory at Kyoto

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15:30-16:30

Title: Sparse Actuation in Linear Dynamical Systems: Controllability and Design Algorithms

Speaker: Geethu Joseph (Delft University of Technology, the Netherlands)



Abstract: This talk explores sparsity-based approaches for controllability analysis and control design in large-scale dynamical systems. As modern systems become increasingly complex, achieving effective control with limited sensing, actuation, and computational resources presents fundamental challenges. We begin by addressing the fundamental question of when sparse control is feasible: under what conditions can a dynamical system be controlled using only a limited number of actuators? Building on these feasibility results, we then investigate how sparse control can be designed and identified through two complementary approaches: sparse actuator scheduling and compressed sensing-based control design. These methods provide systematic ways to determine effective actuator configurations while balancing control performance and resource constraints. Finally, we conclude the talk by briefly discussing several related research directions that I am currently exploring.

Bio: Geethu Joseph received the B. Tech. degree in electronics and communication engineering from the National Institute of Technology, Calicut, India, in 2011, and the M.E. degree in signal

processing and the Ph.D. degree in electrical communication engineering from the Indian Institute of Science, Bangalore, in 2014 and 2019, respectively. She was a postdoctoral fellow with the Department of Electrical Engineering and Computer Science at Syracuse University, NY, USA, from 2019 to 2021. She is currently a tenured assistant professor in the Signal Processing Systems group at the Delft University of Technology, Delft, the Netherlands. Dr. Joseph was awarded the 2022 IEEE SPS Best PhD Dissertation Award and the 2020 SPCOM Best Doctoral Dissertation Award. Dr. Joseph holds 70+ peer-reviewed publications in the fields of signal processing, communications, and control theory. She is an associate editor of the IEEE Sensors Journal and an active reviewer for major journals and conferences in signal processing, communications, and control theory. Her research interests include statistical signal processing, network control, and machine learning.